BUREAU OF INDIAN STANDARDS

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व्यापक परिचालन में मसौदे

प्रलेख प्रेषण संज्ञापन

संदर्भ	दिनाँक
ईटीडी 28/ **टी - 2, टी -3, टी -7, टी -17, टी -40, टी -41** और **टी -42**	30-09-2016

तकनीकी समिति : ईटीडी 28

प्रेषती :

1. ईटीडी 28 के सभी सदस्य
2. विद्युत तकनीकी विभाग परिषद के सभी सदस्य तथा
3. रूचि रखने वाले अन्य सभी निकाय

महोदय,

कृप्या निम्नलिखित मसौदे की एक प्रति संलग्न है :

प्रलेख न.	*शीर्षक*
ईटीडी 28 (10802)	प्रकाशवोल्टीय युक्तियाँ भाग 2 संदर्भ सौर युक्तियों की अपेक्षाएँ (IS 12762 (भाग 2) का दूसरा पुनरीक्षण)
ईटीडी 28 (10803)	प्रकाशवोल्टीय युक्तियाँ भाग 3 संदर्भ वर्णक्रमीय अभिकिरण डाटा सहित स्थलीय प्रकाशवोल्टीय(पी. वी.) सौर युक्ति के लिए मापन सिद्धांत (IS 12762 (भाग 3) का दूसरा पुनरीक्षण)
ईटीडी 28 (10804)	प्रकाशवोल्टीय युक्तियाँ भाग 8 प्रकाशवोल्टीय(पी. वी.) की वर्णक्रमीय प्रतिक्रिया का मापन (IS 12762 (भाग 3) का पहला पुनरीक्षण)
ईटीडी 28 (10805)	यूटिलिटी-इंटरकनेक्टिड इन्वर्टरों हेतु आईलैंडिंग रोकथाम उपायों की परीक्षण पद्धति (IS 16169 का पहला पुनरीक्षण)
ईटीडी 28 (10806)	प्रकाशवोल्टीय प्रणालियों में डी सी अनुप्रयोग हेतु संयोजक - सुरक्षा अपेक्षाएँ एंव परीक्षण
ईटीडी 28 (10807)	प्रकाशवोल्टीय इन्वर्टर - विवरण तालिका एंव नाम पट्टिकाएँ
ईटीडी 28 (10808)	यूटिलिटी-इंटरकनेक्टिड इन्वर्टरों हेतु निम्न वोल्टता राइड थ्रू (एल वी आर टी) मापन हेतु परीक्षण पद्धति

कृप्या इन मसोदों का अवलोकन करें और अपनी सम्मतियॉ यह बताते हुए भेजें कि अंतत: यदि यह मानक के रूप मे प्रकाशित हो जाए तो इस पर अमल करने में आपके व्यवसाय अथवा कारोबार में क्या कठिनाइयॉ आ सकती हैं ।

सम्मतियॉ भेजने की अंतिम तारीख:**30** नवंबर **2016.**

सम्मतियॉ यदि कोई हो तो कृप्या अगले पृष्ठ पर दिए पत्र में अधोहस्ताक्षरी को उपरिलिखित पते पर भेज दें ।यदि कोई सम्मति प्राप्त नहीं होती अथवा सम्मति में केवल भाषा संबंधी त्रुटि हुई तो उपरोक्त प्रलेख को यथावत अतिंम रूप दिया जाएगा । यदि कोई सम्मति तकनीकी प्रकृति की हुई तो विषय समिति के अध्यक्ष के परामर्श से अथवा उनकी इच्छा पर आगे की कार्यवाही के लिए विषय समिति को भेजे जाने के बाद प्रलेख को अतिंम रूप दे दिया जाएगा

धन्यवाद,

भवदीय,

(पी. वी. मैथ्यू)
वैज्ञानिक 'ई' एवं प्रमुख (विद्युत तकनीकी)
संलग्न : उपरिलिखित

BUREAU OF INDIAN STANDARDS

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DRAFTS IN WIDE CIRCULATION

DOCUMENT DESPATCH ADVICE

Reference	Date
ETD 28 T- 2, T-3, T-7, T-17, T-40, T-41 & T-42	30 -09-2016

TECHNICAL COMMITTEE ETD 28

ADDRESSED TO:

1. All Members of Solar Photovoltaic Energy Systems Sectional Committee, ETD 28
2. All Members of Electrotechnical Division Council; and
3. All other Interested.

Dear Sir(s),

Please find enclosed a copy of the following draft Indian Standard:

Please find enclosed a copy of the following draft Indian Standard:

Doc No.	Title
ETD 28(10802)	Photovoltaic devices – Part 2: Requirements for photovoltaic reference devices (Second Revision of IS 12762 (Part 2))
ETD 28(10803)	Photovoltaic devices – Part 3: Measurement principles for terrestrial photovoltaic (PV) solar devices with reference spectral irradiance data(Second Revision of IS 12762 (Part 3))
ETD 28(10804)	Photovoltaic devices – Part 8: Measurement of spectral responsivity of a photovoltaic (PV) device (First Revision of IS 12762 (Part 8))
ETD 28(10805)	Utility-interconnected photovoltaic inverters – Test procedure of islanding prevention measures(First Revision of IS 16169(2014))
ETD 28(10806)	Connectors for DC-application in photovoltaic systems – Safety requirements and tests
ETD 28(10807)	Photovoltaic inverters – Data sheet and name plate
ETD 28(10808)	Utility-interconnected photovoltaic inverters – Test procedure for low voltage ride-through measurements

Kindly examine the draft standards and forward your views stating any difficulties which you are likely to experience in your business or profession, if these are finally adopted as Indian Standards.
Comments, if any, may please be made in the format given overleaf and mailed to the undersigned.
Last date for comments: **30 November 2016**.
In case no comments are received or comments received are of editorial nature, you will kindly permit us to presume your approval for the above document as finalized. However, in case of comments of technical in nature are received then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action, if so desired by the Chairman, Sectional Committee.

Thanking you,

Yours faithfully

(P.V. Mathew)
Sc 'E' & Head (Electrotechnical)
Email: eetd@bis.org.in
Encl : See attachment.

BUREAU OF INDIAN STANDARDS

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Date	Document No.
29-09-2016	Doc: ETD 28 ( )

Sl. No.	Name of the Organization	Clause/ Sub-clause	Paragraph/ Figure/Table	Type of Comment (General/ Technical/ Editorial	Comments	Proposed Change

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY
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Draft Indian Standard

Photovoltaic devices
Part 2 : Requirements for photovoltaic reference devices
(Second Revision)

प्रकाशवोल्टीय युक्तियाँ
भाग 2 संदर्भ सौर युक्तियों की अपेक्षाएँ
(दूसरा पुनरीक्षण)

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Draft Indian Standard(Part 2) (Second Revision) which is identical with IEC 60904-2 : 2015 Photovoltaic devices – Part 2: Requirements for photovoltaic reference devices issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council.

This standard was first published in 1993 and subsequently revised in 2013. The second revision of this standard has been undertaken to align it with the latest version of IEC 60904-2 : 2015 under dual numbering system to make pace with the latest developments that have taken place at International level.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
b) Comma (,) has been used as a decimal marker, while in Indian Standards the current practice is to use a point (.) as the decimal marker.

In this adopted draft standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their respective places, are listed below along with their degree of equivalence for the editions indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60891 Photovoltaic devices – Procedures for temperature and irradiance corrections to measured I-V characteristics	IS 12763 : 2013 Photovoltaic devices — Procedures for temperature and irradiance corrections to measured I-V characteristics (first revision)	Identical to IEC 60891 : 2009
IEC 60904-1 Photovoltaic devices – Part 1 : Measurements of photovoltaic current-voltage characteristics	IS 12762 (Part 1) : 2010 Photovoltaic devices: Part 1 Measurements of photovoltaic current-voltage characteristics (first revision)	Identical to IEC 60904-1 : 2006
IEC 60904-3 Photovoltaic devices – Part 3 : Measurement principles for terrestrial photovoltaic (PV) solar devices with reference spectral irradiance data	IS 12762 (Part 3) : 2013 Photovoltaic Devices: Part 3 Measurement Principles for Terrestrial Photovoltaic (PV) Solar Devices with Reference Spectral Irradiance Data (Under Revision as Doc ETD 28 (10803))	Identical to IEC 60904-3 : 2008
IEC 60904-4 Photovoltaic devices – Part 4 : Reference solar devices – Procedures for establishing calibration traceability	IS 12762(Part 4) : 2014 Photovoltaic Devices: Part 4 Reference Solar Devices Procedures for Establishing Calibration Traceability	Identical to IEC 60904-4 : 2009
IEC 60904-5 Photovoltaic devices – Part 5 : Determination of the equivalent cell temperature (ECT) of photovoltaic (PV) devices by the open-circuit voltage method	IS 12762 (Part 5) : 2010 Photovoltaic devices: Part 5 Determination of the equivalent cell temperature (ECT) of photovoltaic (PV) devices by the open-circuit voltage method	Identical to IEC 60904-5 : 2011
IEC 60904-7 Photovoltaic devices – Part 7 : Computation of the spectral mismatch correction for measurements of photovoltaic devices	IS 12762 (Part 7) : 2013 Photovoltaic devices: Part 7 Computation of the spectral mismatch correction for measurements of photovoltaic devices	Identical to IEC 60904-7 : 2008
IEC 60904-8 Photovoltaic devices – Part 8 : Measurement of spectral responsivity of a photovoltaic (PV) device	IS 12762 (Part 8) : 2010 Photovoltaic devices: Part 8 Measurement of spectral response of a photovoltaic (PV) device (Under Revision as Doc ETD 28 (10804))	Identical to IEC 60904-8 : 1998
IEC 60904-9 Photovoltaic devices – Part 9 : Solar simulator performance requirements	IS 12762 (Part 9) : 2010 Photovoltaic devices: Part 9 Solar simulator performance requirements	Identical to IEC 60904-9:2007
IEC 60904-10 Photovoltaic devices – Part 10 : Methods of linearity measurement	IS 12762 (Part 10) : 2010 Photovoltaic devices: Part 10 Methods of linearity measurement	Identical to IEC 60904-10 : 2009

IEC TS 61836 Solar photovoltaic energy systems – Terms, definitions and symbols	IS 12834 : 2013 Solar photovoltaic energy systems-Terms, Definitions and Symbols	Identical to IEC/TS 61836 : 2007

Only the English language text has been retained while adopting it in this Indian Standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 60904-2 : 2015 or kindly contact:

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Doc: ETD 28 (10803)
(Second Revision of IS 12762 (Part 3))

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY
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Draft Indian Standard

Photovoltaic Devices – Part 3 : Measurement Principles
For Terrestrial Photovoltaic(PV) Solar Devices With Reference Spectral Irradiance Data
(Second Revision)

प्रकाशवोल्टीय युक्तियाँ
भाग 3 संदर्भ वर्णक्रमीय अभिकिरण डाटा सहित स्थलीय प्रकाशवोल्टीय(पी. वी.)
सौर युक्ति के लिए मापन सिद्धांत (दूसरा पुनरीक्षण)

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Indian Standard(Part 3) (Second Revision) which is identical with IEC 60904-3 : 2016 Photovoltaic devices – Part 3 : Measurement principles for terrestrial photovoltaic(PV) solar devices with reference spectral irradiance data issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

This standard was first published in 1998 and subsequently revised in 2013. The second revision of this standard has been undertaken to align it with the latest version of IEC 60904-3 : 2016 under dual numbering system to make pace with the latest developments that have taken place at International level.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
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In this adopted draft standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their respective places, are listed below along with their degree of equivalence for the editions indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60891 Photovoltaic devices – Procedures for temperature and irradiance corrections to measured I-V characteristics	IS 12763 : 2013 Photovoltaic devices — Procedures for temperature and irradiance corrections to measured I-V characteristics (first revision)	Identical to IEC 60891 : 2009
IEC 60904-1 Photovoltaic devices – Part 1 : Measurements of photovoltaic current-voltage characteristics	IS 12762 (Part 1) : 2010 Photovoltaic devices: Part 1 Measurements of photovoltaic current-voltage characteristics (first revision)	Identical to IEC 60904-1 : 2006
IEC 60904-2 Photovoltaic devices – Part 2 : Requirements For Reference Solar Devices	IS 12762 (Part 2) : 2013 Photovoltaic Devices: Part 2 Requirements For Reference Solar Devices (Under Revision as doc no. ETD 28(10802)	Identical to IEC 60904-3 : 2008
IEC 60904-5 Photovoltaic devices – Part 5 : Determination of the equivalent cell temperature (ECT) of photovoltaic (PV) devices by the open-circuit voltage method	IS 12762 (Part 5) : 2010 Photovoltaic devices: Part 5 Determination of the equivalent cell temperature (ECT) of photovoltaic (PV) devices by the open-circuit voltage method	Identical to IEC 60904-5 : 2011
IEC 60904-7 Photovoltaic devices – Part 7 : Computation of the spectral mismatch correction for measurements of photovoltaic devices	IS 12762 (Part 7) : 2013 Photovoltaic devices: Part 7 Computation of the spectral mismatch correction for measurements of photovoltaic devices	Identical to IEC 60904-7 : 2008
IEC 60904-8 Photovoltaic devices – Part 8 : Measurement of spectral responsivity of a photovoltaic (PV) device	IS 12762 (Part 8) : 2010 Photovoltaic devices: Part 8 Measurement of spectral response of a photovoltaic (PV) device (Under Revision as doc no. ETD 28(10804)	Identical to IEC 60904-8 : 1998

Only the English language text has been retained while adopting it in this Indian Standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 60904-3 : 2016 or kindly contact:

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BUREAU OF INDIAN STANDARDS

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Draft Indian Standard

Photovoltaic Devices –Part 8 : Measurement of Spectral Responsivity
of A Photovoltaic (PV) Device
(First Revision)

प्रकाशवोल्टीय युक्तियाँ

भाग 8 प्रकाशवोल्टीय(पी. वी.) युक्ति की वर्णक्रमीय प्रतिक्रिया का मापन

(पहला पुनरीक्षण)

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Indian Standard (Part 8) (First Revision) which is identical with IEC 60904-8 : 2014 Photovoltaic Devices –Part 8 : Measurement of Spectral Responsivity of A Photovoltaic (PV) Device issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

This standard was first published in 2010. The first revision of this standard has been undertaken to align it with the latest version of IEC 60904-8 : 2014 under dual numbering system to make pace with the latest developments that have taken place at International level.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

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In this adopted draft standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their respective places, are listed below along with their degree of equivalence for the editions indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60904-3 Photovoltaic devices – Part 3 : Measurement principles for terrestrial photovoltaic (PV) solar devices with reference spectral irradiance data	IS 12762 (Part 3) : 2013 Photovoltaic Devices: Part 3 Measurement Principles for Terrestrial Photovoltaic (PV) Solar Devices with Reference Spectral Irradiance Data (Under Revision as Doc ETD 28(10803))	Identical to IEC 60904-3 : 2008
IEC 60904-7 Photovoltaic devices – Part 7 : Computation of the spectral mismatch correction for measurements of photovoltaic devices	IS 12762 (Part 7) : 2013 Photovoltaic devices: Part 7 Computation of the spectral mismatch correction for measurements of photovoltaic devices	Identical to IEC 60904-7 : 2008
IEC 60904-9 Photovoltaic devices – Part 9 : Solar simulator performance requirements	IS 12762 (Part 9) : 2010 Photovoltaic devices: Part 9 Solar simulator performance requirements	Identical to IEC 60904-9:2007
IEC TS 61836 Solar photovoltaic energy systems – Terms, definitions and symbols	IS 12834 : 2013 Solar photovoltaic energy systems-Terms, Definitions and Symbols	Identical to IEC/TS 61836 : 2007
IEC 61215 Crystalline silicon terrestrial photovoltaic (PV) modules – Design qualification and type approval	IS 14286:2010 Crystalline Silicon Terrestrial Photovoltaic (PV) modules - Design Qualification And Type Approval	Identical to IEC 61215 : 2005
IEC 61646 Thin-film terrestrial photovoltaic (PV) modules – Design qualification and type approval	IS 16077:2013 Thin-Film Terrestrial Photovoltaic (PV) Modules - Design Qualification and Type Approval	Identical to IEC 61646 : 2008
ISO/IEC 17025 General requirements for the competence of testing and calibration laboratories	IS/ISO/IEC 17025:2005 General requirements for the competence of testing and calibration laboratories	Identical to ISO/IEC 17025 : 2005

Only the English language text has been retained while adopting it in this Indian Standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 60904-8 : 2014 or kindly contact:
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email: eetd@bis.org.in
Telephone: 011-23231192

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY
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Draft Indian Standard
Utility-Interconnected Photovoltaic Inverters –
Test Procedure of Islanding Prevention Measures
(First Revision)

यूटिलिटी-इंटरकनेक्टिड इन्वर्टरों हेतु आईलैडिंग रोकथाम उपायों की परीक्षण पद्धति
(प्रथम पुनरीक्षण)

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Indian Standard (First Revision) which is identical with IEC 62116 : 2014 Utility-interconnected photovoltaic inverters – Test procedure of islanding prevention measures Device issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

This standard was first published in 2014. The first revision of this standard has been undertaken to align it with the latest version of IEC 62116 : 2014 under dual numbering system to make pace with the latest developments that have taken place at International level.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

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International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC TS 61836 Solar photovoltaic energy systems – Terms, definitions and symbols	IS 12834 : 2013 Solar photovoltaic energy systems-Terms, Definitions and Symbols	Identical to IEC/TS 61836 : 2007

Only the English language text has been retained while adopting it in this Indian Standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 62116 : 2014 or kindly contact:

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BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY
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Draft Indian Standard
Connectors for DC-application in photovoltaic systems – Safety requirements and tests

प्रकाशवोल्टीय प्रणालियों में डी सी अनुप्रयोग हेतु संयोजक - सुरक्षा अपेक्षाएँ एंव परीक्षण

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Indian Standard which is identical with IEC 62852 : 2014 Connectors for DC-application in photovoltaic systems – Safety requirements and tests issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

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In this adopted draft standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards, which are to be substituted in their respective places, are listed below along with their degree of equivalence for the editions indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60050 (all parts) : International Electrotechnical Vocabulary (available at http://www.electropedia.org)	IS 1885 (All Parts) Electrotechnical Vocabulary	Technically Equivalent

IEC 60060-1:2010 High-voltage test techniques – Part 1: General definitions and test requirements	IS 2017 : 2016 High-voltage Test Techniques Part 1 General Definitions and Test Requirements (Third Revision)	Identical to IEC 60060-1 : 2010
IEC 60068-1 : 2013 Environmental testing – Part 1: General and guidance	IS 9000(Part 1) : 1988 Basic Environmental Testing Procedures for Electronic and Electrical Items - Part 1 : General	Technically Equivalent to IEC Pub 68-1 (1982)
IEC 60068-2-14 : 2009 Environmental testing – Part 2-14: Tests – Test N: Change of temperature	IS 9000(Part 14/Sec1 to 3):1988 Basic environmental testing procedures for electronic and electrical items:Part 14 Test N:Change of temp	Technically Equivalent to IEC Pub 68-2-14 (1984)
IEC 60068-2-75 : 1997 Environmental testing – Part 2-75: Tests – Test Eh: Hammer tests	IS 9000 (Part 7 / Sec 7) : 2006 Basic Environmental testing procedures for electronic and electrical items Part 7 Impact Test Test Eh Hammer tests	Identical with IEC 60068-2-75 (1997)
IEC 60068-2-78 : 2012 Environmental testing – Part 2-78: Tests – Test Cab : Damp heat, steady state	Basic Environmental Testing Procedures for Electronic and Electrical Items - Part 4 : Damp heat (Steady state)	Identical with IEC 60068-2-78 : 2001
IEC 60228 : 2004 Conductors of insulated cables	IS 8130:2013 Conductors for Insulated Electric Cables and Flexible Cords	Technically Equivalent
IEC 60309-1 : 1999 Plugs, socket-outlets and couplers for industrial purposes – Part 1 : General requirements	IS/IEC 60309-1:2002 Plugs, Socket-Outlets and Couplers for Industrial Purposes - Part 1 : General Requirements	Identical to IEC 60309-1 : 1999
IEC 60529 : 1989 Degrees of protection provided by enclosures (IP Code)	IS / IEC 60529 : 1989 Degrees of protection provided by enclosures (IP Code)	Identical
IEC 60664-1 : 2007 Insulation coordination for equipment within low voltage systems – Part 1 : Principles, requirements and tests	IS 15382(Part 1):2014Insulation Coordination for Equipment Within Low-Voltage Systems Part 1 Principles, Requirements and Tests	Identical to IEC 60664-1 : 2007
IEC 61032 : 1997 Protection of persons and equipment by enclosures – Probes for verification	IS 1401:2008 Protection Of Persons And Equipment By Enclosures - Probes For Verification	Identical to IEC 61032 : 1997
IEC 61215 : 2005 Crystalline silicon terrestrial photovoltaic (PV) modules – Design qualification and type approval	IS 14286:2010 Crystalline Silicon Terrestrial Photovoltaic (PV) modules - Design Qualification And Type	Identical to IEC 61215 : 2005

	Approval	
IEC TS 62548 Photovoltaic (PV) arrays – Design requirements	ETD 28 (06756) Photovoltaic PV arrays Design requirements (Under Development)	Identical

The technical committee responsible for the preparation of this standard has reviewed the provisions of the following International Standards referred to in this adopted standard and has decided that they are acceptable for use in conjunction with this standard:

International Standard	*Title*
IEC 60352-2 : 2006	Solderless connections – Part 2: Solderless crimped connections – General requirements, test methods and practical guidance
IEC 60352-3 : 1993	Solderless connections – Part 3: Solderless accessible insulation displacement connections – General requirements, test methods and practical guidance
IEC 60352-4 : 1994	Solderless connections – Part 4: Solderless non-accessible insulation displacement connections – General requirements, test methods and practical guidance
IEC 60352-5 : 2012	Solderless connections – Part 5 : Press-in connections – General requirements, test methods and practical guidance
IEC 60352-6 : 1997	Solderless connections – Part 6 : Insulation piercing connections – General requirements, test methods and practical guidance
IEC 60352-7 : 2002	Solderless connections – Part 7 : Spring clamp connections – General requirements, test methods and practical guidance
IEC 60364-7-712 : 2002	Electrical installations of buildings – Part 7-712 : Requirements for special installations or locations – Solar photovoltaic (PV) power supply systems
IEC 60512 (all parts)	Connectors for electronic equipment – Tests and measurements
IEC 60512-1 : 2001	Connectors for electronic equipment – Tests and measurements – Part 1 : General
IEC 60512-11-7 : 2003	Electromechanical components for electronic equipment – Basic testing procedures and measuring methods – Part 11-7 : Climatic tests – Test 11g : Flowing mixed gas corrosion test
IEC 60695-2-11 : 2014	Fire hazard testing – Part 2-11: Glowing/hot-wire based test methods - Glow-wire flammability test method for end-products (GWEPT)
IEC 60695-11-10 : 2013	Fire hazard testing – Part 11-10: Test flames – 50 W horizontal and vertical flame test methods
IEC TR 60943 : 1998	Guidance concerning the permissible temperature rise for parts of electrical equipment, in particular for terminals

IEC 60998-2-3 : 2002	Connecting devices for low-voltage circuits for household and similar purposes – Part 2-3 : Particular requirements for connecting devices as separate entities with insulation-piercing clamping units
IEC 60999-1 : 1999	Connecting devices – Electrical copper conductors – Safety requirements for screw-type and screwless-type clamping units – Part 1: General requirements and particular requirements for clamping units for conductors from 0,2 mm2 up to 35 mm2 (included)
IEC 60999-2 : 2003	Connecting devices – Electrical copper conductors – Safety requirements for screw-type and screwless-type clamping units – Part 2: Particular requirements for clamping units for conductors above 35 mm2 up to 300 mm2 (included)
IEC 61140 : 2001	Protection against electric shock – Common aspects for installation and equipment
IEC 61210 : 2010	Connecting devices – Flat quick-connect terminations for electrical copper conductors –Safety requirements
IEC 61984 : 2008	Connectors – Safety requirements and tests
IEC 62444 : 2010	Cable glands for electrical installations
ISO 4892-2	Plastics – Methods of exposure to laboratory light sources – Part 2: Xenon-arc sources
ISO 4892-3	Plastics – Methods of exposure to laboratory light sources – Part 3: Fluorescent UV-lamps
ISO 6988 : 1985	Metallic and other non organic coatings – Sulfur dioxide test with general condensation of moisture

Only the English language text has been retained while adopting it in this Indian Standard and as such the page numbers given here are not the same as in the IEC publication.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 62852 : 2014 or kindly contact:

Head Electrotechnical Department
Bureau of Indian Standards 9, B.S. Zafar Marg, New Delhi-110002
email: eetd@bis.org.in
Telephone: 011-23231192

BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY
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Draft Indian Standard
Photovoltaic inverters – Data sheet and name plate
प्रकाशवोल्टीय इन्वर्टर - विवरण तालिका एंव नाम पट्टिकाएँ

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Indian Standard which is identical with IEC 62894 : 2014 Photovoltaic inverters – Data sheet and name plate issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.
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International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60529 Degrees of protection provided by enclosures (IP Code)	IS/IEC 60529 : 2011 Degrees of protection provided by enclosures (IP Code)	Identical to IEC 60529 : 2011

IEC 60664-1 Insulation coordination for equipment within low-voltage systems – Part 1 : Principles, requirements and tests	IS 15382(Part 1):2014Insulation Coordination for Equipment Within Low-Voltage Systems Part 1 Principles, Requirements and Tests	Identical to IEC 60664-1 : 2007
IEC 60721-2-1 Classification of environmental conditions – Part 2-1 : Environmental conditions appearing in nature – Temperature and humidity	IS 13736(Part 2/Sec 1) : 1993 Classification of environmental conditions: Part 2 Environmental conditions appearing in nature, Sec 1 Temperature and humidity	Identical to IEC 721-2-1 (1982)
IEC 62109-1 Safety of power converters for use in photovoltaic power systems – Part 1 : General requirements	IS 16221(Part 1) : 2016 Safety of Power Converters for use in Photovoltaic Power Systems Part 1 General Requirements	Identical to IEC 62109-1 : 2010
ISO 216 Writing paper and certain classes of printed matter – Trimmed sizes – A and B series, and indication of machine direction	IS 1064 : 1980 Paper Standard Sizes	Technically Equivalent

Only the English language text has been retained while adopting it in this Indian Standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 62894 : 2014 or kindly contact:

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BUREAU OF INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY
(Not to be reproduced without the permission of BIS or used as a STANDARD)

Draft Indian Standard
Utility-Interconnected Photovoltaic Inverters – Test Procedure
For Low Voltage Ride-Through Measurements

यूटिलिटी-इंटरकनेक्टिड इन्वर्टरों हेतु निम्न वोल्टता राइड थ्रू (एल वी आर टी)
मापन हेतु परीक्षण पद्धति

Last date of receipt of comments: 30 November 2016

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This Indian Standard which is identical with IEC 62910 : 2015 Utility-interconnected photovoltaic inverters – Test procedure for low voltage ride-through measurements issued by the International Electrotechnical Commission (IEC)) is proposed to be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain terminology and conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

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International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 61400-21 : 2008 Wind turbines – Part 21 : Measurement and assessment of power quality characteristics of grid connected wind turbines	IS/IEC 61400-21 : 2008 Wind turbines – Part 21 : Measurement and assessment of power quality characteristics of grid connected wind turbines	Identical to IEC 61400-21 : 2008

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding of numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note: The technical content of the document has not been enclosed as these are identical with the corresponding IEC standards. For details, please refer the corresponding IEC 62910 : 2015 or kindly contact:

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